

①

$$\int_0^{\pi/4} \frac{x dx}{1 + \cos 2x + \sin 2x}$$

②

$$\int e^x \left[\frac{1}{(1+x^2)^{3/2}} + \frac{x}{\sqrt{1+x^2}} \right] dx$$

③

Find the coordinates of foot of perpendicular drawn from the point $P(0, 2, 3)$ to the line

$$\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$$

④

$$\int_0^{2\pi} \frac{1}{1 + e^{\sin x}} dx$$

⑤

$$\int \frac{x^4}{(x-1)(x^2+1)} dx$$

⑥

$$\int_0^{\pi/2} \sin 2x \tan^{-1}(\sin x) dx$$

⑦

If $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, then $A^{2023} = ?$

⑧

Find the particular solution of the differential equation $x \frac{dy}{dx} - y = x^2 \cdot e^x$, given that $y(1) = 0$

⑨

$$\int \frac{1}{x^2 - 6x + 13} dx$$

⑩

$$\int_1^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{4-x}} dx$$

⑪

$$\int_{-1}^2 |x^3 - x| dx$$

⑫

Find $\text{adj } A$, if

$$A = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$$